



V2 -- Nema 17 Peristaltic Pump / Water Pump / measuring Pump

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Summary

V2 Peristaltic Pump. Allows you to pump specific fluids without ever touching anything besides the tube itself.

[Hobby & Makers](#) > [Electronics](#)

Peristaltic Pump (NEMA 17)

This is a **compact, fully 3D-printable peristaltic pump** designed to be driven by a **NEMA 17 stepper motor**.

The pump uses standard **608 bearings** and soft tubing, making it easy to build with commonly available parts.

Peristaltic pumps are ideal for liquid handling because the fluid only contacts the tubing. This makes the design suitable for water, coolants, and other non-aggressive liquids.

⚙️ How it works

The stepper motor rotates the bearing rotor.

As the bearings roll over the tubing, they compress it sequentially, pushing the liquid forward.

This design is **self-priming** and naturally prevents backflow without additional valves.

The flow rate can be precisely controlled by adjusting the stepper motor speed.

Bill of Materials

Hardware

- **5× 608 bearings**
- **1× NEMA 17 stepper motor**
- **4× M3 screws (16 mm)** – motor mounting
- **4× M3 screws (15-30 mm)** – housing assembly

Tubing

- **8 mm soft tubing** (silicone recommended)
Slightly smaller or larger tubing also works if the spacer is adjusted accordingly.

Electronics

- Stepper motor driver (A4988, DRV8825, etc.)
- Microcontroller (Arduino or similar)
- Suitable power supply

Print Settings

Bearing holder / rotor

- Infill: **60%**
- Layer height: 0.2 mm
- Material: PLA / PETG / ABS

Housing

- Infill: **15%**
- Layer height: 0.2 mm

Supports are **not required** when printed in the recommended orientation.

Assembly Notes

- Press the **608 bearings** firmly into the rotor (no glue required)
- Insert the tubing into the housing before final assembly
- **Lightly gluing the tubing is recommended** to prevent slipping during operation
- Mount the pump assembly to the stepper motor using the 16 mm M3 screws
- Test the pump at low speed before increasing RPM

⚠ Important Notes

- Use **soft tubing only** – hard tubing will not work
- Not designed for high-pressure applications
- Chemical compatibility depends on the tubing material used

Updates

- **09.04.24** – Improved bearing press-fit based on community feedback
- **10.01.26** – Version 2 files uploaded
- **19.01.26** – Previously missing file added

This remix is based on



V1 -- Nema 17 Peristaltic Pump / Water Pump / measuring Pump

by Maximilian Puschmann

Model files



main-housing.stl



tube-pressure-part.stl



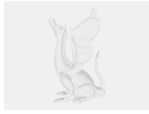
deckel.stl



bearing-place-holder.stl



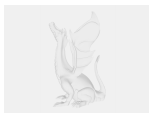
tube-preasure-part.step



v2-main-housing.step



v2-lid.step



peristaltic-pump-v2-finished-model-not-for-printing.step



bearing-place-holder.step

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